1 Continuous-Time ESKF Error-State Derivation

1.1 Nominal state and IMU motion model

Define the nominal state

$$x = \begin{bmatrix} p & v & R & b_g & b_a & g \end{bmatrix}^T.$$

The continuous-time IMU motion equations are

$$\dot{p} = v, \tag{1.1}$$
$$\dot{v} = R(\tilde{a} - b_a - \eta_a) + g, \tag{1.2}$$
$$\dot{R} = R(\tilde{\omega} - b_g - \eta_g)^\wedge, \tag{1.3}$$
$$\dot{b}_g = \eta_{bg}, \tag{1.4}$$
$$\dot{b}_a = \eta_{ba}, \tag{1.5}$$
$$\dot{g} = 0. \tag{1.6}$$

The IMU measurement equations may be written as

$$\tilde{a} = R^T(a - g) + b_a + \eta_a, \tag{1.7}$$
$$\tilde{\omega} = \omega + b_g + \eta_g, \tag{1.8}$$

together with

$$\dot{R} = R\omega^\wedge.$$

For the nominal state, the kinematic equations have the same form but ignore the measurement-noise terms:

$$\dot{p} = v, \tag{1.9}$$
$$\dot{v} = R(\tilde{a} - b_a) + g, \tag{1.10}$$
$$\dot{R} = R(\tilde{\omega} - b_g)^\wedge, \tag{1.11}$$
$$\dot{b}_g = 0, \tag{1.12}$$
$$\dot{b}_a = 0, \tag{1.13}$$
$$\dot{g} = 0. \tag{1.14}$$

1.2 True state

Define the true state as

$$x_t = \begin{bmatrix} p_t & v_t & R_t & b_{at} & b_{gt} & g_t \end{bmatrix}^T.$$

Its continuous-time dynamics are

$$\dot{p}_t = v_t, \tag{1.15}$$
$$\dot{v}_t = R_t(\tilde{a} - b_{at} - \eta_a) + g_t, \tag{1.16}$$
$$\dot{R}_t = R_t(\tilde{\omega} - b_{gt} - \eta_g)^\wedge, \tag{1.17}$$
$$\dot{b}_{gt} = \eta_{bg}, \tag{1.18}$$
$$\dot{b}_{at} = \eta_{ba}, \tag{1.19}$$
$$\dot{g}_t = 0. \tag{1.20}$$

1.3 Definition of the error state

Use a right-multiplicative rotation error:

$$p_t = p + \delta p, \tag{1.21}$$
$$v_t = v + \delta v, \tag{1.22}$$
$$R_t = R\,\delta R = R\,\mathrm{Exp}(\delta\theta), \tag{1.23}$$
$$b_{gt} = b_g + \delta b_g, \tag{1.24}$$
$$b_{at} = b_a + \delta b_a, \tag{1.25}$$
$$g_t = g + \delta g. \tag{1.26}$$

The quantities without the subscript t are nominal-state variables. The goal is to derive the continuous-time dynamics of

$$\delta x = \begin{bmatrix}\delta p & \delta v & \delta\theta & \delta b_g & \delta b_a & \delta g\end{bmatrix}^T.$$

1.4 Position error

Starting from

$$p_t = p + \delta p,$$

differentiate with respect to time:

$$\dot{p}_t = \dot{p} + \delta\dot{p}.$$

Since

$$\dot{p}_t = v_t = v + \delta v$$

and

$$\dot{p} = v,$$

we obtain

$$v + \delta v = v + \delta\dot{p}.$$

Therefore

$$\boxed{\delta\dot{p} = \delta v.} \tag{1.27}$$

1.5 Velocity error

Start from

$$v_t = v + \delta v.$$

Differentiating,

$$\dot{v}_t = \dot{v} + \delta\dot{v}.$$

Using the true-state equation,

$$\dot{v}_t = R_t(\tilde{a} - b_{at} - \eta_a) + g_t.$$

Substitute the error-state definitions:

$$\dot{v}_t = R\,\mathrm{Exp}(\delta\theta)(\tilde{a} - b_a - \delta b_a - \eta_a) + g + \delta g.$$

For small $\delta\theta$,

$$\mathrm{Exp}(\delta\theta) = I + \delta\theta^{\wedge} + O(\|\delta\theta\|^2).$$

Keeping only first-order error terms,

$$\dot{v}_t \approx R(I + \delta\theta^\wedge)(\tilde{a} - b_a - \delta b_a - \eta_a) + g + \delta g \tag{1.28}$$

$$\approx R\tilde{a} - Rb_a - R\delta b_a - R\eta_a + R\delta\theta^\wedge\tilde{a} - R\delta\theta^\wedge b_a + g + \delta g. \tag{1.29}$$

Terms such as

$$\delta\theta^\wedge\delta b_a, \qquad \delta\theta^\wedge\eta_a$$

are neglected as higher-order terms in this first-order error-state linearization.

Using

$$a^\wedge b = -b^\wedge a,$$

we have

$$\delta\theta^\wedge\tilde{a} = -\tilde{a}^\wedge\delta\theta, \qquad -\delta\theta^\wedge b_a = b_a^\wedge\delta\theta.$$

Hence

$$\dot{v}_t \approx R(\tilde{a} - b_a) + g - R(\tilde{a} - b_a)^\wedge\delta\theta - R\delta b_a - R\eta_a + \delta g.$$

On the other hand, the nominal equation gives

$$\dot{v} + \delta\dot{v} = R(\tilde{a} - b_a) + g + \delta\dot{v}.$$

Comparing the two expressions,

$$\boxed{\delta\dot{v} = -R(\tilde{a} - b_a)^\wedge\delta\theta - R\delta b_a - \eta_a + \delta g.} \tag{1.30}$$

Here we use the standard ESKF noise assumption adopted in the text: multiplying the zero-mean white accelerometer noise by a rotation does not change its distribution. Thus, in distribution,

$$R\eta_a \sim \eta_a,$$

and the rotated noise term is written simply as

$$R\eta_a \;\Rightarrow\; \eta_a.$$

This is a distributional simplification rather than a pointwise identity between individual noise samples.

1.6 Rotation error

Start from

$$R_t = R\,\mathrm{Exp}(\delta\theta).$$

Differentiating,

$$\dot{R}_t = \dot{R}\,\mathrm{Exp}(\delta\theta) + R\frac{d}{dt}\,\mathrm{Exp}(\delta\theta).$$

Exact derivative of the exponential map

For a general time-varying

$$\delta\theta(t) \in \mathbb{R}^3,$$

the exact derivative is

$$\boxed{\frac{d}{dt}\operatorname{Exp}(\delta\theta) = \operatorname{Exp}(\delta\theta)\left(J_r(\delta\theta)\delta\dot{\theta}\right)^{\wedge},} \tag{1.31}$$

where $J_r(\delta\theta)$ is the right Jacobian of SO(3).

The formula above can be seen directly from the limit definition of the right Jacobian.

Define

$$Q(t) = \operatorname{Exp}(\delta\theta(t)).$$

The right/local tangent velocity of $Q(t)$ is

$$\omega_{\text{local}} = \lim_{h\to 0} \frac{\operatorname{Log}\big(Q(t)^{-1}Q(t+h)\big)}{h}.$$

Since

$$Q(t)^{-1} = \operatorname{Exp}(-\delta\theta(t))$$

and

$$Q(t+h) = \operatorname{Exp}(\delta\theta(t+h)),$$

we have

$$\omega_{\text{local}} = \lim_{h\to 0} \frac{\operatorname{Log}[\operatorname{Exp}(-\delta\theta(t))\operatorname{Exp}(\delta\theta(t+h))]}{h}.$$

Using the ordinary first-order expansion

$$\delta\theta(t+h) = \delta\theta(t) + h\,\delta\dot{\theta}(t) + O(h^2),$$

this becomes

$$\omega_{\text{local}} = \lim_{h\to 0} \frac{\operatorname{Log}\Big[\operatorname{Exp}(-\delta\theta)\operatorname{Exp}\Big(\delta\theta + h\,\delta\dot{\theta}\Big)\Big]}{h}.$$

Now compare this with the directional-limit definition of the right Jacobian:

$$\boxed{J_r(\phi)u = \lim_{h\to 0} \frac{\operatorname{Log}[\operatorname{Exp}(-\phi)\operatorname{Exp}(\phi + hu)]}{h}.} \tag{1.32}$$

By identifying

$$\phi = \delta\theta, \qquad u = \delta\dot{\theta},$$

we obtain

$$\boxed{\omega_{\text{local}} = J_r(\delta\theta)\delta\dot{\theta}.} \tag{1.33}$$

Since a right/local tangent velocity satisfies

$$\dot{Q} = Q\,\omega_{\text{local}}^{\wedge},$$

we get

$$\dot{Q} = Q\left(J_r(\delta\theta)\delta\dot{\theta}\right)^{\wedge}.$$

Substituting

$$Q = \mathrm{Exp}(\delta\theta)$$

gives the exact relation

$$\boxed{\frac{d}{dt}\,\mathrm{Exp}(\delta\theta) = \mathrm{Exp}(\delta\theta)\left(J_r(\delta\theta)\delta\dot{\theta}\right)^{\wedge}.} \tag{1.34}$$

Equivalently, this is the same relation as the compact right-Jacobian increment formula

$$\mathrm{Exp}(\delta\theta + \delta\dot{\theta}\,dt) \approx \mathrm{Exp}(\delta\theta)\,\mathrm{Exp}\Big(J_r(\delta\theta)\delta\dot{\theta}\,dt\Big).$$

To see why these two forms are equivalent, define

$$Q(t) = \mathrm{Exp}(\delta\theta(t)).$$

Starting from the differential form,

$$\dot{Q} = Q\left(J_r(\delta\theta)\delta\dot{\theta}\right)^{\wedge},$$

the ordinary first-order Taylor expansion in time gives

$$Q(t+dt) = Q(t) + \dot{Q}(t)\,dt + O(dt^2).$$

Substituting the differential equation,

$$Q(t+dt) = Q(t) + Q(t)\left(J_r(\delta\theta)\delta\dot{\theta}\right)^{\wedge} dt + O(dt^2).$$

Factoring out $Q(t)$,

$$Q(t+dt) = Q(t)\left[I + \left(J_r(\delta\theta)\delta\dot{\theta}\right)^{\wedge} dt\right] + O(dt^2).$$

For a small vector ϵ,

$$\mathrm{Exp}(\epsilon) = I + \epsilon^{\wedge} + O(\|\epsilon\|^2).$$

Choose

$$\epsilon = J_r(\delta\theta)\delta\dot{\theta}\,dt.$$

Then

$$\mathrm{Exp}\Big(J_r(\delta\theta)\delta\dot{\theta}\,dt\Big) = I + \left(J_r(\delta\theta)\delta\dot{\theta}\right)^{\wedge} dt + O(dt^2).$$

Therefore

$$Q(t+dt) \approx Q(t)\,\mathrm{Exp}\Big(J_r(\delta\theta)\delta\dot{\theta}\,dt\Big).$$

Since

$$Q(t) = \mathrm{Exp}(\delta\theta(t))$$

and

$$\delta\theta(t+dt) = \delta\theta(t) + \delta\dot{\theta}(t)\,dt + O(dt^2),$$

we obtain

$$\boxed{\mathrm{Exp}(\delta\theta + \delta\dot{\theta}\,dt) \approx \mathrm{Exp}(\delta\theta)\,\mathrm{Exp}\Big(J_r(\delta\theta)\delta\dot{\theta}\,dt\Big).} \tag{1.35}$$

Conversely, start from the increment form:

$$Q(t+dt) \approx Q(t)\,\mathrm{Exp}\Big(J_r(\delta\theta)\delta\dot{\theta}\,dt\Big).$$

Using the small-angle expansion of the second exponential,

$$Q(t+dt) = Q(t) + Q(t)\left(J_r(\delta\theta)\delta\dot{\theta}\right)^{\wedge} dt + O(dt^2).$$

Hence

$$\frac{Q(t+dt)-Q(t)}{dt} = Q(t)\left(J_r(\delta\theta)\delta\dot{\theta}\right)^{\wedge} + O(dt).$$

Taking the limit $dt \to 0$ gives

$$\boxed{\dot{Q} = Q\left(J_r(\delta\theta)\delta\dot{\theta}\right)^{\wedge}.} \tag{1.36}$$

Thus the increment form and the differential form describe the same local motion:

$$\boxed{\begin{array}{c}\mathrm{Exp}(\delta\theta+\delta\dot{\theta}\,dt) \approx \mathrm{Exp}(\delta\theta)\,\mathrm{Exp}\Big(J_r(\delta\theta)\delta\dot{\theta}\,dt\Big)\\ \Updownarrow\ dt\to 0\\ \dfrac{d}{dt}\mathrm{Exp}(\delta\theta) = \mathrm{Exp}(\delta\theta)\left(J_r(\delta\theta)\delta\dot{\theta}\right)^{\wedge}.\end{array}} \tag{1.37}$$

The upper expression is a first-order finite-increment relation, while the lower expression is the corresponding exact instantaneous differential relation.

Since

$$J_r(\delta\theta) = I - \frac{1}{2}\delta\theta^{\wedge} + O(\|\delta\theta\|^2),$$

the first-order ESKF approximation is

$$J_r(\delta\theta)\delta\dot{\theta} \approx \delta\dot{\theta}.$$

Therefore, within the first-order error-state derivation,

$$\boxed{\frac{d}{dt}\mathrm{Exp}(\delta\theta) \approx \mathrm{Exp}(\delta\theta)\delta\dot{\theta}^{\wedge}.} \tag{1.38}$$

Using this first-order approximation,

$$\dot{R}_t \approx \dot{R}\,\mathrm{Exp}(\delta\theta) + R\,\mathrm{Exp}(\delta\theta)\delta\dot{\theta}^{\wedge}.$$

The nominal rotation dynamics are

$$\dot{R} = R(\tilde{\omega} - b_g)^{\wedge}.$$

Hence

$$\dot{R}_t \approx R(\tilde{\omega} - b_g)^{\wedge}\,\mathrm{Exp}(\delta\theta) + R\,\mathrm{Exp}(\delta\theta)\delta\dot{\theta}^{\wedge}.$$

The true rotation dynamics are

$$\dot{R}_t = R_t(\tilde{\omega} - b_{gt} - \eta_g)^{\wedge} = R\,\mathrm{Exp}(\delta\theta)(\tilde{\omega} - b_{gt} - \eta_g)^{\wedge}.$$

Equating both expressions and canceling R gives

$$\mathrm{Exp}(\delta\theta)\delta\dot{\theta}^{\wedge} = \mathrm{Exp}(\delta\theta)(\tilde{\omega} - b_{gt} - \eta_g)^{\wedge} \tag{1.39}$$

$$- (\tilde{\omega} - b_g)^{\wedge}\,\mathrm{Exp}(\delta\theta). \tag{1.40}$$

Use the SO(3) identity

$$\phi^{\wedge}R = R(R^T\phi)^{\wedge}.$$

With

$$R = \mathrm{Exp}(\delta\theta),$$

this gives

$$(\tilde{\omega} - b_g)^{\wedge}\,\mathrm{Exp}(\delta\theta) = \mathrm{Exp}(\delta\theta)\,(\mathrm{Exp}(-\delta\theta)(\tilde{\omega} - b_g))^{\wedge}\,.$$

Therefore

$$\mathrm{Exp}(\delta\theta)\delta\dot{\theta}^{\wedge} = \mathrm{Exp}(\delta\theta)\Big[(\tilde{\omega} - b_{gt} - \eta_g)^{\wedge} \tag{1.41}$$

$$- (\mathrm{Exp}(-\delta\theta)(\tilde{\omega} - b_g))^{\wedge}\Big]. \tag{1.42}$$

For small $\delta\theta$,

$$\mathrm{Exp}(-\delta\theta) \approx I - \delta\theta^{\wedge}.$$

Thus

$$(\tilde{\omega} - b_{gt} - \eta_g) - \mathrm{Exp}(-\delta\theta)(\tilde{\omega} - b_g) \tag{1.43}$$

$$\approx (\tilde{\omega} - b_{gt} - \eta_g) - (I - \delta\theta^{\wedge})(\tilde{\omega} - b_g) \tag{1.44}$$

$$= b_g - b_{gt} - \eta_g + \delta\theta^{\wedge}\tilde{\omega} - \delta\theta^{\wedge}b_g. \tag{1.45}$$

Since

$$b_{gt} = b_g + \delta b_g,$$

we have

$$b_g - b_{gt} = -\delta b_g.$$

Also,

$$\delta\theta^{\wedge}(\tilde{\omega} - b_g) = -(\tilde{\omega} - b_g)^{\wedge}\delta\theta.$$

Therefore

$$\delta\dot{\theta} = -(\tilde{\omega} - b_g)^{\wedge}\delta\theta - \delta b_g - \eta_g.$$

Hence

$$\boxed{\delta\dot{\theta} = -(\tilde{\omega} - b_g)^{\wedge}\delta\theta - \delta b_g - \eta_g.} \tag{1.46}$$

Identity used above

For $R \in \mathrm{SO}(3)$,

$$\mathrm{Ad}_R\,\phi = (R\phi^{\wedge}R^T)^{\vee} = R\phi.$$

Thus

$$R\phi^{\wedge}R^T = (R\phi)^{\wedge}.$$

From this,

$$\phi^{\wedge}R = R(R^T\phi)^{\wedge}.$$

1.7 Supplement: From Finite Infinitesimal Increments to Instantaneous Velocity

This appendix clarifies how the right-Jacobian finite-increment relation is used to obtain the instantaneous right/local tangent velocity.

Let

$$Q(t) = \mathrm{Exp}(\delta\theta(t)).$$

For a small time increment dt,

$$\delta\theta(t+dt) = \delta\theta(t) + \delta\dot{\theta}(t)\,dt + O(dt^2),$$

and therefore

$$Q(t+dt) = \mathrm{Exp}(\delta\theta(t+dt)) \approx \mathrm{Exp}(\delta\theta + \delta\dot{\theta}\,dt).$$

The right-Jacobian increment relation is

$$\boxed{\mathrm{Exp}(\phi + \Delta\phi) \approx \mathrm{Exp}(\phi)\,\mathrm{Exp}(J_r(\phi)\Delta\phi)\,.} \tag{1.47}$$

Set

$$\phi = \delta\theta, \qquad \Delta\phi = \delta\dot{\theta}\,dt.$$

Then

$$\boxed{Q(t+dt) \approx Q(t)\,\mathrm{Exp}\Big(J_r(\delta\theta)\delta\dot{\theta}\,dt\Big)\,.} \tag{1.48}$$

Now define the instantaneous right/local tangent velocity ω_R by the infinitesimal motion

$$\boxed{Q(t+dt) \approx Q(t)\,\mathrm{Exp}(\omega_R\,dt).} \tag{1.49}$$

Comparing the two infinitesimal expressions gives

$$\boxed{\omega_R = J_r(\delta\theta)\delta\dot{\theta}.} \tag{1.50}$$

Thus $\delta\dot{\theta}$ is the time derivative of the exponential coordinates, whereas $J_r(\delta\theta)\delta\dot{\theta}$ is the corresponding right/local tangent velocity of the rotation matrix.

To recover the ordinary matrix derivative, use

$$\mathrm{Exp}(\omega_R\,dt) = I + \omega_R^{\wedge}dt + O(dt^2).$$

Hence

$$Q(t+dt) = Q(t) + Q(t)\omega_R^{\wedge}dt + O(dt^2),$$

so

$$\frac{Q(t+dt) - Q(t)}{dt} = Q(t)\omega_R^{\wedge} + O(dt).$$

Taking $dt \to 0$,

$$\boxed{\dot{Q} = Q\omega_R^{\wedge}.} \tag{1.51}$$

Substituting

$$\omega_R = J_r(\delta\theta)\delta\dot{\theta}$$

and

$$Q = \mathrm{Exp}(\delta\theta)$$

gives

$$\boxed{\frac{d}{dt}\operatorname{Exp}(\delta\theta) = \operatorname{Exp}(\delta\theta)\left(J_r(\delta\theta)\delta\dot{\theta}\right)^{\wedge}.} \tag{1.52}$$

Therefore the finite infinitesimal increment and the instantaneous differential form are two descriptions of the same local motion:

$$\boxed{\begin{array}{c} Q(t+dt) \approx Q(t)\operatorname{Exp}\Big(J_r(\delta\theta)\delta\dot{\theta}\,dt\Big) \\ \Downarrow \quad dt \to 0 \\ \dot{Q} = Q\left(J_r(\delta\theta)\delta\dot{\theta}\right)^{\wedge}. \end{array}} \tag{1.53}$$

Exact and small-angle forms of the right Jacobian

For $\phi \in \mathbb{R}^3$, let

$$\vartheta = \|\phi\|.$$

The exact right Jacobian of SO(3) is

$$\boxed{J_r(\phi) = I - \frac{1-\cos\vartheta}{\vartheta^2}\phi^{\wedge} + \frac{\vartheta - \sin\vartheta}{\vartheta^3}(\phi^{\wedge})^2.} \tag{1.54}$$

The coefficient of the first-order term follows from

$$\cos\vartheta = 1 - \frac{\vartheta^2}{2!} + \frac{\vartheta^4}{4!} - \cdots,$$

so

$$\frac{1-\cos\vartheta}{\vartheta^2} = \frac{1}{2} - \frac{\vartheta^2}{24} + \cdots.$$

Similarly,

$$\sin\vartheta = \vartheta - \frac{\vartheta^3}{3!} + \frac{\vartheta^5}{5!} - \cdots,$$

which gives

$$\frac{\vartheta - \sin\vartheta}{\vartheta^3} = \frac{1}{6} - \frac{\vartheta^2}{120} + \cdots.$$

Therefore

$$J_r(\phi) = I - \frac{1}{2}\phi^{\wedge} + \frac{1}{6}(\phi^{\wedge})^2 + O(\|\phi\|^3),$$

and, retaining only first-order terms,

$$\boxed{J_r(\phi) = I - \frac{1}{2}\phi^{\wedge} + O(\|\phi\|^2).} \tag{1.55}$$

For the ESKF rotation error, $\phi = \delta\theta$ is small. Hence

$$J_r(\delta\theta)\delta\dot{\theta} = \delta\dot{\theta} - \frac{1}{2}\delta\theta^{\wedge}\delta\dot{\theta} + \cdots.$$

Under the first-order error-state approximation, the product involving $\delta\theta$ and $\delta\dot{\theta}$ is neglected, yielding

$$\boxed{J_r(\delta\theta)\delta\dot{\theta} \approx \delta\dot{\theta}.} \tag{1.56}$$

This is why the ESKF derivation commonly uses

$$\frac{d}{dt}\operatorname{Exp}(\delta\theta) \approx \operatorname{Exp}(\delta\theta)\delta\dot{\theta}^{\wedge},$$

even though the exact differential relation contains the right Jacobian.

1.8 Gyroscope-bias error

Starting from

$$b_{gt} = b_g + \delta b_g,$$

differentiate:

$$\dot{b}_{gt} = \dot{b}_g + \delta\dot{b}_g.$$

The true bias dynamics are

$$\dot{b}_{gt} = \eta_{bg},$$

while the nominal model ignores noise:

$$\dot{b}_g = 0.$$

Therefore

$$\boxed{\delta\dot{b}_g = \eta_{bg}.} \tag{1.57}$$

1.9 Accelerometer-bias error

Similarly,

$$b_{at} = b_a + \delta b_a.$$

Differentiating,

$$\dot{b}_{at} = \dot{b}_a + \delta\dot{b}_a.$$

Using

$$\dot{b}_{at} = \eta_{ba}, \qquad \dot{b}_a = 0,$$

we obtain

$$\boxed{\delta\dot{b}_a = \eta_{ba}.} \tag{1.58}$$

1.10 Gravity error

Starting from

$$g_t = g + \delta g,$$

differentiate:

$$\dot{g}_t = \dot{g} + \delta\dot{g}.$$

Since

$$\dot{g}_t = 0, \qquad \dot{g} = 0,$$

we obtain

$$\boxed{\delta\dot{g} = 0.} \tag{1.59}$$

2 Final continuous-time error-state equations

Collecting the results,

$$\boxed{\begin{aligned}
\delta\dot{p} &= \delta v, \\
\delta\dot{v} &= -R(\tilde{a} - b_a)^{\wedge}\delta\theta - R\delta b_a - \eta_a + \delta g, \\
\delta\dot{\theta} &= -(\tilde{\omega} - b_g)^{\wedge}\delta\theta - \delta b_g - \eta_g, \\
\delta\dot{b}_g &= \eta_{bg}, \\
\delta\dot{b}_a &= \eta_{ba}, \\
\delta\dot{g} &= 0.
\end{aligned}} \tag{2.1}$$

3 Discrete-Time ESKF Kinematic Equation

Notation convention

In the continuous-time derivations below, t, τ, and s denote physical time. After discretization, the final propagation equations use the discrete time-step indices k and $k+1$, with

$$\boxed{t_k = t, \qquad t_{k+1} = t_k + \Delta t = t + \Delta t.} \tag{3.1}$$

Thus, continuous-time integration uses t, τ, s, while final discrete-time propagation equations use $k, k+1$.

3.1 Nominal-state position equation

For the nominal state,

$$\dot{p} = v, \qquad \dot{v} = R(\tilde{a} - b_a) + g.$$

Define the acceleration expressed in the world frame as

$$\boxed{a_W(t) = R(t)(\tilde{a}(t) - b_a(t)) + g.} \tag{3.2}$$

Since

$$\dot{p}(\tau) = v(\tau),$$

integrating from t to $t + \Delta t$ gives

$$p(t + \Delta t) = p(t) + \int_t^{t+\Delta t} v(\tau)\, d\tau.$$

The velocity at an intermediate time τ is

$$v(\tau) = v(t) + \int_t^{\tau} a_W(s)\, ds.$$

Substituting this into the position integral,

$$p(t + \Delta t) = p(t) + \int_t^{t+\Delta t} \left[v(t) + \int_t^{\tau} a_W(s)\, ds \right] d\tau.$$

Assumption over one IMU sampling interval

To obtain the discrete-time equation used here, assume that the world-frame acceleration is approximately constant over the short IMU sampling interval:

$$\boxed{a_W(s) \approx a_W(t), \qquad s \in [t, t+\Delta t].} \tag{3.3}$$

Equivalently, over this interval we use the values at the beginning of the interval,

$$R(s) \approx R(t), \qquad \tilde{a}(s) \approx \tilde{a}(t), \qquad b_a(s) \approx b_a(t),$$

with gravity g treated as constant. Thus

$$a_W(s) \approx R(t)(\tilde{a}(t) - b_a(t)) + g.$$

Under this approximation,

$$\int_t^\tau a_W(s)\, ds \approx a_W(t)(\tau - t).$$

Therefore

$$p(t+\Delta t) = p(t) + \int_t^{t+\Delta t} [v(t) + a_W(t)(\tau - t)]\, d\tau \tag{3.4}$$

$$= p(t) + v(t)\Delta t + a_W(t) \int_t^{t+\Delta t} (\tau - t)\, d\tau. \tag{3.5}$$

The remaining integral is

$$\int_t^{t+\Delta t} (\tau - t)\, d\tau = \frac{1}{2}\Delta t^2.$$

Hence

$$p(t+\Delta t) = p(t) + v(t)\Delta t + \frac{1}{2} a_W(t)\Delta t^2.$$

Substituting the expression for $a_W(t)$,

$$p(t+\Delta t) = p(t) + v(t)\Delta t + \frac{1}{2}\left[R(t)(\tilde{a}(t) - b_a(t)) + g\right]\Delta t^2.$$

Finally,

$$\boxed{p_{k+1} = p_k + v_k \Delta t + \frac{1}{2} R_k(\tilde{a}_k - b_{a,k})\Delta t^2 + \frac{1}{2} g_k \Delta t^2.} \tag{3.6}$$

This is the discrete-time nominal position propagation equation. The appearance of $R(t)$ rather than $R(t+\Delta t)$ follows from the beginning-of-interval approximation above. In the exact continuous motion, $R(s)$ and therefore $a_W(s)$ generally vary throughout the sampling interval.

3.2 Nominal-state velocity equation

Starting from

$$\dot{v} = R(\tilde{a} - b_a) + g \equiv a_W,$$

integrate over the interval $[t, t+\Delta t]$:

$$v(t+\Delta t) = v(t) + \int_t^{t+\Delta t} a_W(\tau)\, d\tau.$$

Using the same assumption as in the position derivation,

$$a_W(\tau) \approx a_W(t), \qquad \tau \in [t, t + \Delta t],$$

we obtain

$$v(t + \Delta t) \approx v(t) + a_W(t) \int_t^{t+\Delta t} d\tau.$$

Therefore

$$v(t + \Delta t) = v(t) + a_W(t)\Delta t.$$

Substituting

$$a_W(t) = R(t)(\tilde{a}(t) - b_a(t)) + g,$$

gives

$$\boxed{v_{k+1} = v_k + R_k(\tilde{a}_k - b_{a,k})\Delta t + g_k \Delta t.} \tag{3.7}$$

This is the discrete-time nominal velocity propagation equation. As with the position equation, the approximation uses the beginning-of- interval values $R(t)$, $\tilde{a}(t)$, and $b_a(t)$ over one IMU sampling interval.

3.3 Nominal-state rotation equation

The continuous-time nominal rotation dynamics are

$$\boxed{\dot{R}(t) = R(t)(\tilde{\omega}(t) - b_g(t))^\wedge.} \tag{3.8}$$

Define the bias-corrected body-frame angular velocity

$$\omega_B(t) \equiv \tilde{\omega}(t) - b_g(t).$$

Over one short IMU sampling interval, assume

$$\boxed{\omega_B(\tau) \approx \omega_B(t), \qquad \tau \in [t, t + \Delta t].} \tag{3.9}$$

Equivalently,

$$\tilde{\omega}(\tau) \approx \tilde{\omega}(t), \qquad b_g(\tau) \approx b_g(t)$$

over this interval. The differential equation therefore becomes

$$\dot{R}(\tau) = R(\tau)\omega_B(t)^\wedge,$$

where $\omega_B(t)^\wedge$ is constant with respect to τ.

The solution of this matrix differential equation with initial condition $R(t)$ is

$$R(\tau) = R(t) \exp\big(\omega_B(t)^\wedge(\tau - t)\big),$$

where $\exp(\cdot)$ denotes the matrix exponential.

This can be verified directly:

$$\frac{d}{d\tau}R(\tau) = R(t)\frac{d}{d\tau}\exp\big(\omega_B(t)^\wedge(\tau - t)\big) \tag{3.10}$$

$$= R(t)\exp\big(\omega_B(t)^\wedge(\tau - t)\big)\,\omega_B(t)^\wedge \tag{3.11}$$

$$= R(\tau)\omega_B(t)^\wedge. \tag{3.12}$$

At $\tau = t + \Delta t$,

$$R(t + \Delta t) = R(t) \exp\big(\omega_B(t)^\wedge \Delta t\big) .$$

Since

$$\mathrm{Exp}(\phi) \equiv \exp(\phi^\wedge)$$

and

$$(\omega_B \Delta t)^\wedge = \omega_B^\wedge \Delta t,$$

we obtain

$$R(t + \Delta t) = R(t) \,\mathrm{Exp}(\omega_B(t)\Delta t) .$$

Substituting

$$\omega_B(t) = \tilde{\omega}(t) - b_g(t),$$

gives the discrete-time nominal rotation propagation equation

$$\boxed{R_{k+1} = R_k \,\mathrm{Exp}((\tilde{\omega}_k - b_{g,k})\Delta t) .} \tag{3.13}$$

The exponential increment appears on the right because

$$\dot{R} = R\omega_B^\wedge$$

uses the body-frame (right-trivialized) angular velocity. In contrast, if the angular velocity were expressed in the world frame so that

$$\dot{R} = \omega_W^\wedge R,$$

the corresponding constant-angular-velocity propagation would be

$$R(t + \Delta t) = \mathrm{Exp}(\omega_W \Delta t) R(t).$$

3.4 Nominal-state bias and gravity equations

For the nominal state, the gyroscope bias, accelerometer bias, and gravity are modeled as constant during propagation:

$$\dot{b}_g = 0, \qquad \dot{b}_a = 0, \qquad \dot{g} = 0.$$

Integrating the gyroscope-bias equation over one sampling interval,

$$b_g(t + \Delta t) = b_g(t) + \int_t^{t+\Delta t} \dot{b}_g(\tau)\, d\tau,$$

gives

$$\boxed{b_{g,k+1} = b_{g,k}.} \tag{3.14}$$

Similarly,

$$b_a(t + \Delta t) = b_a(t) + \int_t^{t+\Delta t} \dot{b}_a(\tau)\, d\tau,$$

so

$$\boxed{b_{a,k+1} = b_{a,k}.} \tag{3.15}$$

For gravity,

$$g(t + \Delta t) = g(t) + \int_t^{t+\Delta t} \dot{g}(\tau)\, d\tau,$$

and therefore

$$\boxed{g_{k+1} = g_k.} \tag{3.16}$$

The constant nominal-bias equations do not imply that the physical IMU biases are assumed to remain exactly constant. Their random-walk variations are represented in the error-state dynamics by

$$\delta\dot{b}_g = \eta_{bg}, \qquad \delta\dot{b}_a = \eta_{ba},$$

and their uncertainty is propagated through the error-state covariance.

3.5 Summary of the discrete-time nominal-state equations

Collecting the nominal-state propagation equations,

$$\boxed{\begin{aligned} p_{k+1} &= p_k + v_k\Delta t + \frac{1}{2}\left[R_k(\tilde{a}_k - b_{a,k}) + g_k\right]\Delta t^2, \\ v_{k+1} &= v_k + \left[R_k(\tilde{a}_k - b_{a,k}) + g_k\right]\Delta t, \\ R_{k+1} &= R_k \operatorname{Exp}((\tilde{\omega}_k - b_{g,k})\Delta t)\,, \\ b_{g,k+1} &= b_{g,k}, \\ b_{a,k+1} &= b_{a,k}, \\ g_{k+1} &= g_k. \end{aligned}} \tag{3.17}$$

3.6 Discrete-time error-state equations

We now discretize the continuous-time error-state equations derived previously.

Position error

From the continuous-time position-error equation,

$$\delta\dot{p} = \delta v,$$

assume that δv is approximately constant over one short sampling interval $[t, t + \Delta t]$. Then

$$\delta p(t + \Delta t) - \delta p(t) = \int_t^{t+\Delta t} \delta\dot{p}(\tau)\, d\tau \tag{3.18}$$

$$= \int_t^{t+\Delta t} \delta v(\tau)\, d\tau \tag{3.19}$$

$$\approx \delta v(t)\Delta t. \tag{3.20}$$

Therefore,

$$\boxed{\delta p_{k+1} = \delta p_k + \delta v_k \Delta t.} \tag{3.21}$$

Velocity error

From the continuous-time velocity-error equation,

$$\delta\dot{v} = -R(\tilde{a} - b_a)^\wedge \delta\theta - R\delta b_a - \eta_a + \delta g,$$

define the deterministic part

$$\boxed{f_v = -R(\tilde{a} - b_a)^\wedge \delta\theta - R\delta b_a + \delta g.} \tag{3.22}$$

Then

$$\delta\dot{v} = f_v - \eta_a.$$

Integrating over one sampling interval,

$$\delta v(t + \Delta t) - \delta v(t) = \int_t^{t+\Delta t} f_v(\tau)\, d\tau - \int_t^{t+\Delta t} \eta_a(\tau)\, d\tau. \tag{3.23}$$

Assume that the deterministic quantities in f_v are approximately constant over the short interval:

$$\boxed{f_v(\tau) \approx f_v(t), \qquad \tau \in [t, t + \Delta t].} \tag{3.24}$$

Thus

$$\int_t^{t+\Delta t} f_v(\tau)\, d\tau \approx f_v(t)\Delta t.$$

Define the integrated accelerometer-noise increment

$$\boxed{\eta_v \equiv \int_t^{t+\Delta t} \eta_a(\tau)\, d\tau.} \tag{3.25}$$

Assume the continuous-time accelerometer noise is zero-mean white Gaussian noise with

$$\mathbb{E}[\eta_a(t)] = 0$$

and

$$\boxed{\mathbb{E}\big[\eta_a(t)\eta_a(s)^T\big] = \sigma_a^2 I\, \delta(t - s).} \tag{3.26}$$

Equivalently, σ_a^2 is the continuous-time white-noise intensity (per axis). Then the integrated noise increment also has zero mean:

$$\mathbb{E}[\eta_v] = 0.$$

Its covariance is

$$\begin{aligned}
\mathrm{Cov}(\eta_v) &= \mathbb{E}\big[\eta_v \eta_v^T\big] && (3.27)\\
&= \int_t^{t+\Delta t} \int_t^{t+\Delta t} \mathbb{E}\big[\eta_a(\tau)\eta_a(s)^T\big]\, d\tau\, ds && (3.28)\\
&= \sigma_a^2 I \int_t^{t+\Delta t} \int_t^{t+\Delta t} \delta(\tau - s)\, d\tau\, ds && (3.29)\\
&= \boxed{\sigma_a^2 \Delta t\, I.} && (3.30)
\end{aligned}$$

Thus, per axis,

$$\boxed{\mathrm{Var}(\eta_v) = \sigma_a^2 \Delta t.} \tag{3.31}$$

The book distinguishes the sampled accelerometer measurement noise from the continuous-time noise-density parameter through

$$\boxed{\sigma_a(k) = \frac{\sigma_a}{\sqrt{\Delta t}}.} \tag{3.32}$$

Therefore the standard deviation of the integrated velocity-noise increment can be written in either equivalent form:

$$\boxed{\sigma(\eta_v) = \Delta t\, \sigma_a(k) = \sigma_a \sqrt{\Delta t}.} \tag{3.33}$$

The first expression uses the sampled-noise notation, while the second follows directly after substituting the sampled-noise relation.

Note that writing simply

$$\mathrm{Var}(\eta_a) = \sigma_a^2$$

for continuous-time white noise is shorthand for its noise intensity. The Dirac-delta covariance above makes the continuous-to-discrete scaling explicit.

Hence

$$\delta v(t + \Delta t) \approx \delta v(t) + f_v(t)\Delta t - \eta_v.$$

Substituting the expression for $f_v(t)$ gives

$$\boxed{\begin{aligned}\delta v(t + \Delta t) = \delta v(t) &- R(t)(\tilde{a}(t) - b_a(t))^{\wedge}\delta\theta(t)\Delta t \\ &- R(t)\delta b_a(t)\Delta t + \delta g(t)\Delta t - \eta_v.\end{aligned}} \tag{3.34}$$

The term η_v is the discrete-time noise increment obtained by integrating the continuous-time accelerometer noise over the sampling interval; unlike the deterministic term $f_v(t)\Delta t$, it should not be interpreted simply as a deterministic constant multiplied by Δt.

Rotation error

From the continuous-time rotation-error equation,

$$\delta\dot{\theta} = -(\tilde{\omega} - b_g)^{\wedge}\delta\theta - \delta b_g - \eta_g,$$

define

$$A \equiv -(\tilde{\omega} - b_g)^{\wedge}.$$

Over one short IMU sampling interval, assume that

$$\tilde{\omega}(\tau) \approx \tilde{\omega}(t), \qquad b_g(\tau) \approx b_g(t),$$

so that A is approximately constant. The homogeneous equation

$$\dot{x} = Ax$$

has the solution

$$x(t + \Delta t) = \exp(A\Delta t)x(t).$$

Therefore the homogeneous part of the rotation error propagates as

$$\delta\theta(t + \Delta t) = \exp\big(-(\tilde{\omega}(t) - b_g(t))^{\wedge}\Delta t\big)\, \delta\theta(t).$$

Using the SO(3) exponential-map notation,

$$\exp\big(-(\tilde{\omega} - b_g)^{\wedge}\Delta t\big) = \mathrm{Exp}(-(\tilde{\omega} - b_g)\Delta t)\,.$$

For the forcing terms $-\delta b_g - \eta_g$, the first-order discretization used here gives

$$\boxed{\begin{aligned}\delta\theta(t+\Delta t) \approx \operatorname{Exp}(-(\tilde{\omega}(t) - b_g(t))\Delta t)\, \delta\theta(t) \\ - \delta b_g(t)\Delta t - \eta_\theta,\end{aligned}} \tag{3.35}$$

where the integrated gyroscope-noise increment is

$$\boxed{\eta_\theta \equiv \int_t^{t+\Delta t} \eta_g(\tau)\, d\tau.} \tag{3.36}$$

More exactly, for constant

$$A = -(\tilde{\omega} - b_g)^\wedge,$$

the variation-of-constants formula is

$$\delta\theta(t+\Delta t) = e^{A\Delta t}\delta\theta(t) - \int_t^{t+\Delta t} e^{A(t+\Delta t-\tau)}\, [\delta b_g(\tau) + \eta_g(\tau)]\, d\tau.$$

The expression above with $-\delta b_g \Delta t - \eta_\theta$ is the corresponding first-order forcing-term approximation.

Variance of the discrete rotation-noise increment. Assume the continuous-time gyroscope noise is zero-mean isotropic white Gaussian noise with

$$\mathbb{E}[\eta_g(t)] = 0$$

and

$$\boxed{\mathbb{E}\big[\eta_g(t)\eta_g(s)^T\big] = \sigma_g^2 I\, \delta(t-s).} \tag{3.37}$$

Then

$$\mathbb{E}[\eta_\theta] = 0$$

and

$$\operatorname{Cov}(\eta_\theta) = \mathbb{E}\big[\eta_\theta \eta_\theta^T\big] \tag{3.38}$$

$$= \int_t^{t+\Delta t}\int_t^{t+\Delta t} \mathbb{E}\big[\eta_g(\tau)\eta_g(s)^T\big]\, d\tau\, ds \tag{3.39}$$

$$= \boxed{\sigma_g^2 \Delta t\, I.} \tag{3.40}$$

Thus, per axis,

$$\boxed{\operatorname{Var}(\eta_\theta) = \sigma_g^2 \Delta t.} \tag{3.41}$$

For the sampled gyroscope measurement noise, the book uses

$$\boxed{\sigma_g(k) = \frac{\sigma_g}{\sqrt{\Delta t}}.} \tag{3.42}$$

Hence the standard deviation of the integrated rotation-noise increment can be written as

$$\boxed{\sigma(\eta_\theta) = \Delta t\, \sigma_g(k) = \sigma_g \sqrt{\Delta t}.} \tag{3.43}$$

Again, the first form uses the sampled-noise notation, while the second is the equivalent expression in terms of the continuous-time noise-density parameter. The $\sqrt{\Delta t}$ factor belongs to the standard deviation, not to the variance. The same covariance result also holds if the exact state-transition rotation inside the noise integral is retained: because A is skew-symmetric, e^{As} is a rotation matrix, and for isotropic noise

$$e^{As}(\sigma_g^2 I)e^{As\,T} = \sigma_g^2 I.$$

Gyroscope-bias error

Here we explicitly distinguish the continuous-time bias random-walk process from its integrated discrete-time increment. The continuous process is denoted by

$$\eta_{bg}(t),$$

while

$$\eta_{bg,d}$$

denotes the noise accumulated over one sampling interval. The book reuses the symbol η_{bg} for the discrete increment in Eqs. (3.42)–(3.43); the subscript d is retained here to avoid ambiguity.

From

$$\delta\dot{b}_g = \eta_{bg},$$

integrating over one sampling interval gives

$$\delta b_g(t + \Delta t) = \delta b_g(t) + \int_t^{t+\Delta t} \eta_{bg}(\tau)\, d\tau.$$

Define

$$\boxed{\eta_{bg,d} \equiv \int_t^{t+\Delta t} \eta_{bg}(\tau)\, d\tau.} \tag{3.44}$$

Then

$$\boxed{\delta b_{g,k+1} = \delta b_{g,k} + \eta_{bg,d}.} \tag{3.45}$$

If

$$\mathbb{E}\big[\eta_{bg}(t)\eta_{bg}(s)^T\big] = \sigma_{bg}^2 I\, \delta(t - s),$$

then

$$\boxed{\mathrm{Cov}(\eta_{bg,d}) = \sigma_{bg}^2 \Delta t\, I,} \tag{3.46}$$

or, per axis,

$$\boxed{\mathrm{Var}(\eta_{bg,d}) = \sigma_{bg}^2 \Delta t, \qquad \sigma_{\eta_{bg,d}} = \sigma_{bg}\sqrt{\Delta t}.} \tag{3.47}$$

Accelerometer-bias error

Likewise, $\eta_{ba}(t)$ denotes the continuous-time accelerometer-bias random-walk process, while $\eta_{ba,d}$ denotes its integrated discrete-time increment over one sampling interval.

Similarly, from

$$\delta\dot{b}_a = \eta_{ba},$$

we obtain

$$\boxed{\delta b_a(t + \Delta t) = \delta b_a(t) + \eta_{ba,d},} \tag{3.48}$$

where

$$\boxed{\eta_{ba,d} \equiv \int_t^{t+\Delta t} \eta_{ba}(\tau)\, d\tau.} \tag{3.49}$$

For continuous-time isotropic white noise intensity σ_{ba}^2,

$$\boxed{\mathrm{Cov}(\eta_{ba,d}) = \sigma_{ba}^2 \Delta t\, I,} \tag{3.50}$$

and therefore

$$\boxed{\mathrm{Var}(\eta_{ba,d}) = \sigma_{ba}^2 \Delta t, \qquad \sigma_{\eta_{ba,d}} = \sigma_{ba}\sqrt{\Delta t}.} \tag{3.51}$$

Gravity error

From

$$\delta\dot{g} = 0,$$

integration gives

$$\delta g(t+\Delta t) - \delta g(t) = \int_t^{t+\Delta t} \delta\dot{g}(\tau)\, d\tau = 0.$$

Therefore

$$\boxed{\delta g_{k+1} = \delta g_k.} \tag{3.52}$$

3.7 Summary of the discrete-time error-state equations

Collecting the discrete-time error-state propagation equations,

$$\boxed{\begin{aligned}
\delta p_{k+1} &= \delta p_k + \delta v_k \Delta t, \\
\delta v_{k+1} &= \delta v_k - R_k(\tilde{a}_k - b_{a,k})^\wedge \delta\theta_k \Delta t - R_k \delta b_{a,k} \Delta t + \delta g_k \Delta t - \eta_v, \\
\delta\theta_{k+1} &\approx \mathrm{Exp}(-(\tilde{\omega}_k - b_{g,k})\Delta t)\, \delta\theta_k - \delta b_{g,k}\Delta t - \eta_\theta, \\
\delta b_{g,k+1} &= \delta b_{g,k} + \eta_{bg,d}, \\
\delta b_{a,k+1} &= \delta b_{a,k} + \eta_{ba,d}, \\
\delta g_{k+1} &= \delta g_k.
\end{aligned}} \tag{3.53}$$

It is useful to list both the standard deviations and variances. The sampled IMU measurement-noise standard deviations satisfy

$$\sigma_a(k) = \frac{\sigma_a}{\sqrt{\Delta t}}, \qquad \sigma_g(k) = \frac{\sigma_g}{\sqrt{\Delta t}}.$$

Therefore,

$$\boxed{\begin{aligned}
\sigma(\eta_v) &= \Delta t\, \sigma_a(k) = \sigma_a \sqrt{\Delta t}, & \mathrm{Var}(\eta_v) &= \sigma_a^2 \Delta t, \\
\sigma(\eta_\theta) &= \Delta t\, \sigma_g(k) = \sigma_g \sqrt{\Delta t}, & \mathrm{Var}(\eta_\theta) &= \sigma_g^2 \Delta t, \\
\sigma(\eta_{bg,d}) &= \sigma_{bg}\sqrt{\Delta t}, & \mathrm{Var}(\eta_{bg,d}) &= \sigma_{bg}^2 \Delta t, \\
\sigma(\eta_{ba,d}) &= \sigma_{ba}\sqrt{\Delta t}, & \mathrm{Var}(\eta_{ba,d}) &= \sigma_{ba}^2 \Delta t.
\end{aligned}} \tag{3.54}$$

The first two standard-deviation expressions use the sampled IMU-noise notation. For the bias random walks, the book denotes the integrated discrete noises again by η_{bg} and η_{ba}; this note uses $\eta_{bg,d}$ and $\eta_{ba,d}$ so that they remain distinct from the continuous-time processes $\eta_{bg}(t)$ and $\eta_{ba}(t)$.

4 Motion Process of ESKF

Using the discrete-time error-state equations derived above, write the error-state motion model as

$$\boxed{\delta x_{k+1} = f(\delta x_k) + w_k, \qquad w_k \sim \mathcal{N}(0, Q_k).} \tag{4.1}$$

The discrete process-noise covariance is

$$\boxed{Q_k = \mathrm{diag}\left(0_3,\, \mathrm{Cov}(\eta_v),\, \mathrm{Cov}(\eta_\theta),\, \mathrm{Cov}(\eta_{bg,d}),\, \mathrm{Cov}(\eta_{ba,d}),\, 0_3\right).} \tag{4.2}$$

Here the subscript d is retained for the bias-noise increments in order to distinguish them from the corresponding continuous-time white-noise processes.

4.1 Linearized form

Linearize the discrete transition function f about the nominal error-state mean. For a general linearization point $\bar{\delta x}_k$,

$$f(\delta x_k) \approx f(\bar{\delta x}_k) + \left.\frac{\partial f(\delta x)}{\partial \delta x}\right|_{\delta x = \bar{\delta x}_k} (\delta x_k - \bar{\delta x}_k).$$

In the ESKF, the error-state mean is reset to zero, so the linearization point is

$$\bar{\delta x}_k = 0.$$

Therefore,

$$f(\delta x_k) \approx f(0) + \left.\frac{\partial f(\delta x)}{\partial \delta x}\right|_{\delta x = 0} \delta x_k.$$

For the error-state motion model,

$$f(0) = 0.$$

Define the discrete-time error-state transition Jacobian as

$$\boxed{F_k \triangleq \left.\frac{\partial f(\delta x)}{\partial \delta x}\right|_{\delta x = 0}.} \tag{4.3}$$

The linearized error-state propagation is therefore

$$\boxed{\delta x_{k+1} \approx F_k \delta x_k + w_k.} \tag{4.4}$$

Thus, F_k is the Jacobian of the *discrete-time error-state transition function* with respect to the error state, evaluated at zero error. It should not be confused with the time derivative $d\delta x/dt$.

4.2 State transition Jacobian matrix

The error-state ordering is

$$\boxed{\delta x = \begin{bmatrix} \delta p & \delta v & \delta \theta & \delta b_g & \delta b_a & \delta g \end{bmatrix}^T.} \tag{4.5}$$

Since the discrete-time error-state equations have already been linearized explicitly, the transition Jacobian can be read directly from the coefficients multiplying each error-state component. With the ordering above,

$$\boxed{F_k = \begin{bmatrix} I_3 & I_3\Delta t & 0 & 0 & 0 & 0 \\ 0 & I_3 & -R(\tilde{a} - b_a)^\wedge \Delta t & 0 & -R\Delta t & I_3\Delta t \\ 0 & 0 & \mathrm{Exp}(-(\tilde{\omega} - b_g)\Delta t) & -I_3\Delta t & 0 & 0 \\ 0 & 0 & 0 & I_3 & 0 & 0 \\ 0 & 0 & 0 & 0 & I_3 & 0 \\ 0 & 0 & 0 & 0 & 0 & I_3 \end{bmatrix}.} \tag{4.6}$$

Each entry is a 3×3 block, so $F_k \in \mathbb{R}^{18\times 18}$.

4.3 ESKF IMU propagation flow

For each new IMU measurement, the ESKF prediction process can be summarized as follows:

$$
\boxed{
\begin{array}{c}
\text{New IMU measurement} \\
(\tilde{a}_k,\ \tilde{\omega}_k,\ \Delta t) \\
\downarrow \\
\text{Nominal-state prediction} \\
\text{using the discrete-time nominal-state equations} \\
\downarrow \\
x_{k+1}^{\text{pred}} = [p,\ v,\ R,\ b_g,\ b_a,\ g]^T \\
\downarrow \\
\text{Build the error-state transition matrix } F_k \\
\text{using the error-state transition Jacobian} \\
\downarrow \\
\begin{array}{ll}
\delta x_{k+1}^{\text{pred}} = F_k \delta x_k & P_{k+1}^{\text{pred}} = F_k P_k F_k^T + Q_k \\
\text{usually remains zero} & \text{propagate uncertainty}
\end{array} \\
\downarrow \\
\text{Predicted ESKF state} \\
\left(x_{k+1}^{\text{pred}},\ \delta x_{k+1}^{\text{pred}},\ P_{k+1}^{\text{pred}}\right).
\end{array}
} \tag{4.7}
$$

Mathematically, the prediction contains three steps:

1. propagate the nominal state using the IMU measurements;
2. propagate the error-state mean,
$$\delta x_{k+1}^{\text{pred}} = F_k \delta x_k;$$
3. propagate the error-state covariance,
$$P_{k+1}^{\text{pred}} = F_k P_k F_k^T + Q_k.$$

In the ESKF, the error-state mean is reset to zero after the update step. Therefore, in the usual prediction cycle,

$$\delta x_k = 0 \quad \Longrightarrow \quad \delta x_{k+1}^{\text{pred}} = F_k \delta x_k = 0.$$

Consequently, the explicit error-state mean propagation is often unnecessary in an implementation. The important quantities propagated during the IMU prediction are the nominal state and the error-state covariance.

5 Update Process of ESKF

5.1 ESKF update flow

The ESKF update process can be summarized as

New sensor observation z

$\downarrow$

Construct the observation model

$$z = h(x) + v, \qquad v \sim \mathcal{N}(0, V)$$

$\downarrow$

Compute the observation Jacobian H

$$H = \frac{\partial h}{\partial \delta x}$$

$\downarrow$

Compute the innovation

$$r = z - h(x_{\text{pred}})$$

$\downarrow$

Compute the Kalman gain

$$K = P_{\text{pred}} H^T \left(H P_{\text{pred}} H^T + V \right)^{-1}$$

$\downarrow$

Estimate the error state

$$\delta x = Kr$$

$\downarrow$

Update the error-state covariance

$$P = (I - KH) P_{\text{pred}}$$

$\downarrow$

Pass $(\delta x, P)$ to ESKF post-processing

Thus, the observation update itself estimates the posterior error state and its covariance. The estimated error state is not yet the final nominal-state correction at this stage. The following post-processing stage injects δx into the nominal state, resets the error-state mean to zero, and, when retained, transforms the covariance to the new tangent-space reference.

For the full-pose GNSS implementation discussed below, the innovation takes the manifold-aware form

$$r_{\text{GNSS}} = \begin{bmatrix} p_{\text{gnss}} - p \\ \text{Log}(R^T R_{\text{gnss}}) \end{bmatrix},$$

and the direct error-state observation gives

$$H_{\text{GNSS}} = \begin{bmatrix} I_3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & I_3 & 0 & 0 & 0 \end{bmatrix}.$$

5.2 Observation model

Consider the general observation equation

$$\boxed{z = h(x) + v, \qquad v \sim \mathcal{N}(0, V).} \tag{5.1}$$

Here z is the observation data, v is the observation noise, and V is the observation-noise covariance matrix.

In a traditional EKF, the observation equation is linearized with respect to the state variables. In the ESKF, the Kalman-filter update is performed with respect to the error-state variables. Therefore, the required observation Jacobian is

$$\boxed{H = \left.\frac{\partial h}{\partial \delta x}\right|_{x_{\text{pred}}}.} \tag{5.2}$$

5.3 Kalman update equations

Once the observation Jacobian is available, the Kalman gain is

$$\boxed{K = P_{\text{pred}} H^T \left(H P_{\text{pred}} H^T + V\right)^{-1}.} \tag{5.3}$$

The error-state update is

$$\boxed{\delta x = K\left(z - h(x_{\text{pred}})\right).} \tag{5.4}$$

Following the notation used in this chapter, the nominal state is then updated as

$$\boxed{x = x_{\text{pred}} + \delta x.} \tag{5.5}$$

The covariance update is

$$\boxed{P = (I - KH) P_{\text{pred}}.} \tag{5.6}$$

Here K is the Kalman gain, P_{pred} is the predicted error-state covariance, and P is the updated covariance.

5.4 Observation Jacobian with respect to the error state

For observations of the nominal state, the Jacobian with respect to the error state can be obtained using the chain rule:

$$\boxed{H = \frac{\partial h}{\partial x}\frac{\partial x}{\partial \delta x}.} \tag{5.7}$$

The chain of mappings is

$$\boxed{\delta x \longrightarrow x(\delta x) \longrightarrow h(x(\delta x)).}$$

Here $x(\delta x)$ denotes a local perturbation of the nominal state used for linearization. It should not be confused with the true state.

5.5 Local perturbation parameterization

For the nominal-state ordering

$$x = \begin{bmatrix} p & v & R & b_g & b_a & g \end{bmatrix}^T,$$

define the locally perturbed state by

$$\boxed{x(\delta x) = \begin{bmatrix} p + \delta p \\ v + \delta v \\ R\,\mathrm{Exp}(\delta\theta) \\ b_g + \delta b_g \\ b_a + \delta b_a \\ g + \delta g \end{bmatrix}, \qquad x(0) = x_{\mathrm{pred}}.} \tag{5.8}$$

For the Euclidean components,

$$\frac{\partial(p + \delta p)}{\partial \delta p} = I_3, \qquad \frac{\partial(v + \delta v)}{\partial \delta v} = I_3,$$

and similarly

$$\frac{\partial(b_g + \delta b_g)}{\partial \delta b_g} = I_3, \qquad \frac{\partial(b_a + \delta b_a)}{\partial \delta b_a} = I_3, \qquad \frac{\partial(g + \delta g)}{\partial \delta g} = I_3.$$

The rotation component requires special treatment. Since the perturbation is applied on the right,

$$R(\delta\theta) = R\,\mathrm{Exp}(\delta\theta),$$

and the corresponding rotation coordinate is

$$\mathrm{Log}(R\,\mathrm{Exp}(\delta\theta))\,.$$

Therefore,

$$\left.\frac{\partial\,\mathrm{Log}(R\,\mathrm{Exp}(\delta\theta))}{\partial \delta\theta}\right|_{\delta\theta=0} = J_r^{-1}(\mathrm{Log}\,R). \tag{5.9}$$

Consequently,

$$\boxed{\frac{\partial x}{\partial \delta x} = \mathrm{diag}\left(I_3,\, I_3,\, J_r^{-1}(\mathrm{Log}\,R),\, I_3,\, I_3,\, I_3\right).} \tag{5.10}$$

Rotation block: chain-rule derivation

The rotation block can also be derived by explicitly applying the chain rule to the composition

$$\delta\theta \longrightarrow \mathrm{Exp}(\delta\theta) \longrightarrow R\,\mathrm{Exp}(\delta\theta) \longrightarrow \mathrm{Log}(R\,\mathrm{Exp}(\delta\theta))\,.$$

Therefore,

$$\frac{\partial\,\mathrm{Log}(R\,\mathrm{Exp}(\delta\theta))}{\partial \delta\theta} = J^{\mathrm{Log}(R\,\mathrm{Exp}(\delta\theta))}_{R\,\mathrm{Exp}(\delta\theta)}\, J^{R\,\mathrm{Exp}(\delta\theta)}_{\mathrm{Exp}(\delta\theta)}\, J^{\mathrm{Exp}(\delta\theta)}_{\delta\theta}. \tag{5.11}$$

Using the right-trivialized Jacobians,

$$J_Q^{\mathrm{Log}(Q)} = J_r^{-1}(\mathrm{Log}\,Q), \qquad J_Q^{RQ} = I_3, \qquad J_{\delta\theta}^{\mathrm{Exp}(\delta\theta)} = J_r(\delta\theta),$$

we obtain

$$\frac{\partial \operatorname{Log}(R \operatorname{Exp}(\delta\theta))}{\partial \delta\theta} = J_r^{-1}(\operatorname{Log}(R \operatorname{Exp}(\delta\theta)))\, J_r(\delta\theta). \tag{5.12}$$

Evaluating at $\delta\theta = 0$ and using

$$\operatorname{Exp}(0) = I_3, \qquad J_r(0) = I_3,$$

gives

$$\boxed{\left.\frac{\partial \operatorname{Log}(R \operatorname{Exp}(\delta\theta))}{\partial \delta\theta}\right|_{\delta\theta=0} = J_r^{-1}(\operatorname{Log} R).} \tag{5.13}$$

Rotation block: BCH result

Let

$$\phi = \operatorname{Log} R, \qquad R = \operatorname{Exp}(\phi).$$

Using the textbook right-BCH linear approximation,

$$\boxed{\operatorname{Log}(\operatorname{Exp}(\phi) \operatorname{Exp}(\delta\theta)) \approx \phi + J_r^{-1}(\phi)\delta\theta.} \tag{5.14}$$

Since $R = \operatorname{Exp}(\phi)$ and $\phi = \operatorname{Log} R$,

$$\boxed{\left.\frac{\partial \operatorname{Log}(R \operatorname{Exp}(\delta\theta))}{\partial \delta\theta}\right|_{\delta\theta=0} = J_r^{-1}(\operatorname{Log} R).} \tag{5.15}$$

The full BCH-series derivation is collected in the appendix.

5.6 Full-pose GNSS observation

For the full-pose GNSS case, the observation contains position and rotation. With the nominal-state ordering above, the observation function is conceptually

$$\boxed{h(x) = \begin{bmatrix} p \\ R \end{bmatrix}.} \tag{5.16}$$

Thus the GNSS observation directly depends on the nominal position p and nominal rotation R, while it does not directly observe v, b_g, b_a, or g.

The next step is to evaluate

$$\frac{\partial h}{\partial x} = \begin{bmatrix} \frac{\partial h}{\partial p} & \frac{\partial h}{\partial v} & \frac{\partial h}{\partial R} & \frac{\partial h}{\partial b_g} & \frac{\partial h}{\partial b_a} & \frac{\partial h}{\partial g} \end{bmatrix},$$

and then combine it with $\partial x / \partial \delta x$. The treatment of the rotation observation and its relation to the direct error-state GNSS formulation is left for the following discussion.

Two observation formulations

For the full-pose GNSS update, the position observation is straightforward. The rotation observation, however, can be formulated in two different ways.

General chain-rule formulation

If the GNSS rotation is represented in Lie-algebra coordinates, define

$$\phi = \operatorname{Log} R, \qquad \phi_{\text{gnss}} = \operatorname{Log} R_{\text{gnss}}.$$

The nominal-state observation function can then be written as

$$\boxed{h(x) = \begin{bmatrix} p \\ \operatorname{Log} R \end{bmatrix}.} \tag{5.17}$$

With the nominal-state ordering

$$x = \begin{bmatrix} p & v & R & b_g & b_a & g \end{bmatrix}^T,$$

the observation Jacobian with respect to the nominal-state coordinates selects the position and rotation components:

$$\boxed{\frac{\partial h}{\partial x} = \begin{bmatrix} I_3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & I_3 & 0 & 0 & 0 \end{bmatrix}.} \tag{5.18}$$

From the local perturbation derivation above,

$$\frac{\partial x}{\partial \delta x} = \operatorname{diag}\left(I_3,\, I_3,\, J_r^{-1}(\operatorname{Log} R),\, I_3,\, I_3,\, I_3\right).$$

Therefore,

$$\boxed{H_{\text{chain}} = \frac{\partial h}{\partial x}\frac{\partial x}{\partial \delta x} = \begin{bmatrix} I_3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & J_r^{-1}(\operatorname{Log} R) & 0 & 0 & 0 \end{bmatrix}.} \tag{5.19}$$

In this formulation, the rotation innovation is represented in Lie-algebra coordinates as

$$\operatorname{Log} R_{\text{gnss}} - \operatorname{Log} R.$$

Direct error-state observation for GNSS rotation

For the GNSS implementation, a more natural formulation is available because the ESKF rotation error is defined by right multiplication:

$$\boxed{R_{\text{gnss}} = R \operatorname{Exp}(\delta\theta).} \tag{5.20}$$

Left-multiplying by R^T gives

$$R^T R_{\text{gnss}} = \operatorname{Exp}(\delta\theta).$$

Taking the logarithm yields

$$\boxed{z_{\delta\theta} = \operatorname{Log}(R^T R_{\text{gnss}}) = \delta\theta.} \tag{5.21}$$

Thus the GNSS rotation is a direct observation of the rotation error state, and its Jacobian is simply

$$\boxed{\frac{\partial z_{\delta\theta}}{\partial \delta\theta} = I_3.} \tag{5.22}$$

The position observation is similarly

$$p_{\text{gnss}} = p + \delta p,$$

so

$$\frac{\partial p_{\text{gnss}}}{\partial \delta p} = I_3.$$

Consequently, the full-pose GNSS observation Jacobian used in the implementation is

$$\boxed{H_{\text{GNSS}} = \begin{bmatrix} I_3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & I_3 & 0 & 0 & 0 \end{bmatrix}.} \tag{5.23}$$

The corresponding innovation is

$$\boxed{z - h(x_{\text{pred}}) = \begin{bmatrix} p_{\text{gnss}} - p \\ \text{Log}(R^T R_{\text{gnss}}) \end{bmatrix}.} \tag{5.24}$$

Which formulation is used?

The general chain-rule formulation is mathematically valid when the rotation observation is represented using the coordinate $\text{Log}\, R$. In that case, the rotation Jacobian contains $J_r^{-1}(\text{Log}\, R)$.

For the full-pose GNSS update, however, the direct error-state formulation is preferable because

$$\text{Log}(R^T R_{\text{gnss}})$$

is the relative rotation from the nominal estimate to the GNSS observation and is exactly aligned with the right rotation error $\delta\theta$. Therefore the implementation uses

$$\boxed{H_R = I_3}$$

rather than approximating $J_r^{-1}(\text{Log}\, R)$ by the identity. The two forms correspond to different observation representations; the identity block is not obtained by discarding the right-Jacobian term.

5.7 Body-frame 3D Wheel Velocity Observation

Observation model

The wheel encoder directly measures the forward body-frame speed. To form a three-dimensional body-frame observation, we additionally impose approximate zero lateral and vertical body velocities as vehicle-motion constraints:

$$\boxed{z_b = \begin{bmatrix} v_{\text{wheel}} & 0 & 0 \end{bmatrix}^T, \qquad z_b = h(x) + n_b, \qquad n_b \sim \mathcal{N}(0, V_b).} \tag{5.25}$$

The last two components are pseudo-measurements, not independent readings from the wheel encoder. With R mapping body coordinates into world coordinates and v denoting world-frame velocity, the observation function is

$$\boxed{h(x) = R^T v.} \tag{5.26}$$

The ESKF error state is ordered as $\delta x = [\delta p^T, \delta v^T, \delta\theta^T, \delta b_g^T, \delta b_a^T, \delta g^T]^T$.

Velocity Jacobian

With $v(\delta v) = v + \delta v$, keeping R fixed,

$$h(\delta v) = R^T(v + \delta v) = R^T v + R^T \delta v.$$

Consequently,

$$\boxed{\left.\frac{\partial h}{\partial \delta v}\right|_{\delta v = 0} = R^T.} \tag{5.27}$$

Rotation Jacobian: Method 1 — first-order expansion

Using the right perturbation $R(\delta\theta) = R\,\mathrm{Exp}(\delta\theta)$,

$$h(\delta x) = \big(R\,\mathrm{Exp}(\delta\theta)\big)^T(v + \delta v) = \mathrm{Exp}(-\delta\theta)R^T(v + \delta v).$$

The first-order exponential approximation is $\mathrm{Exp}(-\delta\theta) \approx I - \delta\theta^\wedge$. Neglecting the second-order product $\delta\theta^\wedge R^T \delta v$,

$$\begin{aligned} h(\delta x) &\approx (I - \delta\theta^\wedge)(R^T v + R^T \delta v) && (5.28)\\ &\approx R^T v + R^T \delta v - \delta\theta^\wedge R^T v && (5.29)\\ &= R^T v + R^T \delta v + (R^T v)^\wedge \delta\theta. && (5.30) \end{aligned}$$

We used $-a^\wedge b = b^\wedge a$. Thus,

$$\boxed{\left.\frac{\partial h}{\partial \delta\theta}\right|_{\delta\theta = 0} = (R^T v)^\wedge.} \tag{5.31}$$

Rotation Jacobian: Method 2 — directly from the limit definition

Fix the nominal R, v and let $u \in \mathbb{R}^3$ be an arbitrary direction. The directional derivative at zero perturbation is

$$\begin{aligned} Dh(0)[u] &= \lim_{\epsilon \to 0} \frac{\mathrm{Exp}(-\epsilon u)R^T v - R^T v}{\epsilon} && (5.32)\\ &= \left(\lim_{\epsilon \to 0} \frac{\mathrm{Exp}(-\epsilon u) - I}{\epsilon}\right) R^T v && (5.33)\\ &= -u^\wedge R^T v && (5.34)\\ &= (R^T v)^\wedge u. && (5.35) \end{aligned}$$

The limit in the second line equals $-u^\wedge$ by the definition of the differential of the exponential map at the identity. Because u is arbitrary, the linear map is represented by

$$\boxed{\left.\frac{\partial h}{\partial \delta\theta}\right|_{\delta\theta = 0} = (R^T v)^\wedge.} \tag{5.36}$$

Both methods therefore give the same sign and the same rotation block.

Complete error-state observation Jacobian

The observation has no direct dependence on position, IMU biases, or gravity. Collecting the six 3×3 blocks gives

$$\boxed{H_b = \begin{bmatrix} 0_{3\times3} & R^T & (R^T v)^\wedge & 0_{3\times3} & 0_{3\times3} & 0_{3\times3} \end{bmatrix} \in \mathbb{R}^{3\times18}.} \tag{5.37}$$

Measurement covariance and Kalman update

We distinguish the uncertainty of the measured longitudinal wheel speed from the uncertainty of the lateral and vertical motion constraints:

$$\boxed{V_b = \mathrm{diag}(\sigma_x^2, \sigma_y^2, \sigma_z^2).} \tag{5.38}$$

The values σ_y^2 and σ_z^2 express confidence in the pseudo-measurements; they should be relaxed when sideslip, uneven ground, or other violations of the zero-velocity assumptions are plausible.

The innovation and Kalman update are

$$r_b = z_b - R^T v, \tag{5.39}$$
$$K = P_{\text{pred}} H_b^T (H_b P_{\text{pred}} H_b^T + V_b)^{-1}, \tag{5.40}$$
$$\delta x = K r_b, \tag{5.41}$$
$$P = (I - K H_b) P_{\text{pred}}. \tag{5.42}$$

The estimated correction and covariance are then passed to the post-processing stage for nominal-state injection and reset.

Comparison with the textbook's world-frame simplification

The textbook first rotates the wheel-speed vector into world coordinates, $z_w = R v_{\text{wheel}}$, and then treats that converted observation as fixed while using $h_w(x) = v$. This gives

$$H_{\text{book}} = \begin{bmatrix} 0 & I_3 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

Here we instead retain the body-frame measurement model $h(x) = R^T v$, which explicitly accounts for both velocity and rotation perturbations. These are different observation formulations, not an algebraic correction to the textbook. The body-frame model also makes clear that the two zero-valued components are modeling constraints rather than encoder measurements.

6 Post-Processing of ESKF Error State

After the prediction and observation-update processes, the Kalman filter has obtained an estimate of the error state. The remaining steps are to inject this estimated error into the nominal state and then reset the ESKF error state.

6.1 Error-state injection into the nominal state

Let the estimated error-state correction be

$$\delta x_k = \begin{bmatrix} \delta p_k & \delta v_k & \delta \theta_k & \delta b_{g,k} & \delta b_{a,k} & \delta g_k \end{bmatrix}^T.$$

The nominal state is corrected component by component:

$$\boxed{\begin{aligned} p_{k+1} &= p_k + \delta p_k, \\ v_{k+1} &= v_k + \delta v_k, \\ R_{k+1} &= R_k \operatorname{Exp}(\delta\theta_k), \\ b_{g,k+1} &= b_{g,k} + \delta b_{g,k}, \\ b_{a,k+1} &= b_{a,k} + \delta b_{a,k}, \\ g_{k+1} &= g_k + \delta g_k. \end{aligned}} \tag{6.1}$$

The rotation correction is applied through the exponential map rather than ordinary matrix addition.

6.2 Error-state mean reset

Once the estimated error has been injected into the nominal state, the error-state mean is reset:

$$\boxed{\delta x^+ = 0.} \tag{6.2}$$

The estimated error has not been discarded; it has already been absorbed into the corrected nominal state. The reset simply makes the corrected nominal state the new reference point for the next ESKF cycle.

6.3 Covariance reset and change of tangent space

Before the reset, the covariance P describes the error distribution in the tangent space associated with the old nominal state. After the nominal state is corrected, the error state is defined relative to a new reference point. For Euclidean state components this change does not introduce an additional transformation, but the rotation component requires special treatment.

Let R_k be the nominal rotation before injection, and let $\delta\theta_k$ be the estimated rotation correction obtained from the Kalman update. After injection,

$$\boxed{R^+ = R_k \operatorname{Exp}(\delta\theta_k).} \tag{6.3}$$

Let $\delta\theta$ denote a rotation-error random variable expressed relative to the old nominal rotation, and let $\delta\theta^+$ denote the same rotation uncertainty expressed relative to the new nominal rotation. Both descriptions must represent the same physical rotation:

$$\boxed{R^+ \operatorname{Exp}(\delta\theta^+) = R_k \operatorname{Exp}(\delta\theta).} \tag{6.4}$$

Substituting $R^+ = R_k \operatorname{Exp}(\delta\theta_k)$ gives

$$R_k \operatorname{Exp}(\delta\theta_k) \operatorname{Exp}(\delta\theta^+) = R_k \operatorname{Exp}(\delta\theta).$$

Canceling R_k and left-multiplying by $\operatorname{Exp}(-\delta\theta_k)$ yields

$$\boxed{\operatorname{Exp}(\delta\theta^+) = \operatorname{Exp}(-\delta\theta_k) \operatorname{Exp}(\delta\theta).} \tag{6.5}$$

Using the linearized BCH formula for small $\delta\theta$,

$$\boxed{\delta\theta^+ \approx -\delta\theta_k + \delta\theta - \frac{1}{2}\delta\theta_k^{\wedge}\delta\theta.} \tag{6.6}$$

Therefore the reset Jacobian for the rotation-error component is

$$\boxed{J_\theta \triangleq \frac{\partial \delta\theta^+}{\partial \delta\theta} \approx I_3 - \frac{1}{2}\delta\theta_k^\wedge.} \tag{6.7}$$

Extending this transformation to the complete error state gives

$$\boxed{J_k = \operatorname{diag}\left(I_3,\, I_3,\, J_\theta,\, I_3,\, I_3,\, I_3\right).} \tag{6.8}$$

The covariance must therefore be transformed to the tangent space of the corrected nominal state:

$$\boxed{P_{\text{reset}} = J_k P J_k^T.} \tag{6.9}$$

6.4 Interpretation

The post-processing stage can be summarized as

$$\boxed{\begin{array}{c} \text{Kalman update estimates } \delta x_k \\ \downarrow \\ \text{Inject } \delta x_k \text{ into the nominal state} \\ \downarrow \\ \delta x^+ \leftarrow 0 \\ \downarrow \\ P_{\text{reset}} = J_k P J_k^T. \end{array}}$$

The covariance-reset operation is a change of tangent-space reference: the Gaussian error distribution is re-expressed around the corrected nominal state. Since $\delta\theta_k$ is normally small,

$$J_\theta = I_3 - \frac{1}{2}\delta\theta_k^\wedge \approx I_3,$$

so many ESKF implementations neglect this small correction and use $P_{\text{reset}} \approx P$. Nevertheless, the reset Jacobian makes the tangent-space transformation explicit.

A Detailed BCH Derivation for the Rotation Observation Jacobian

For completeness, this appendix derives the right-BCH linear approximation used in the update-process section.

Let

$$\phi = \operatorname{Log} R, \qquad R = \operatorname{Exp}(\phi).$$

Then

$$\operatorname{Log}(R \operatorname{Exp}(\delta\theta)) = \operatorname{Log}(\operatorname{Exp}(\phi) \operatorname{Exp}(\delta\theta)).$$

The BCH series gives

$$\log(e^A e^B) = A + B + \frac{1}{2}[A,B] + \frac{1}{12}[A,[A,B]] + \frac{1}{12}[B,[B,A]] + \cdots.$$

Set

$$A = \phi^\wedge, \qquad B = \delta\theta^\wedge.$$

Keeping only terms that are first order in $\delta\theta$,

$$\log\left(e^{\phi^\wedge} e^{\delta\theta^\wedge}\right) = \phi^\wedge + \delta\theta^\wedge + \frac{1}{2}[\phi^\wedge, \delta\theta^\wedge] + \frac{1}{12}[\phi^\wedge, [\phi^\wedge, \delta\theta^\wedge]] + \cdots + O(\|\delta\theta\|^2).$$

For SO(3),

$$[a^\wedge, b^\wedge] = (a^\wedge b)^\wedge.$$

Therefore,

$$\mathrm{Log}(\mathrm{Exp}(\phi)\,\mathrm{Exp}(\delta\theta)) = \phi + \left[I + \frac{1}{2}\phi^\wedge + \frac{1}{12}(\phi^\wedge)^2 + \cdots\right]\delta\theta + O(\|\delta\theta\|^2).$$

The bracketed series is the inverse right Jacobian,

$$J_r^{-1}(\phi) = I + \frac{1}{2}\phi^\wedge + \frac{1}{12}(\phi^\wedge)^2 + \cdots.$$

Hence

$$\mathrm{Log}(\mathrm{Exp}(\phi)\,\mathrm{Exp}(\delta\theta)) = \phi + J_r^{-1}(\phi)\delta\theta + O(\|\delta\theta\|^2),$$

which gives

$$\boxed{\left.\frac{\partial\,\mathrm{Log}(R\,\mathrm{Exp}(\delta\theta))}{\partial\delta\theta}\right|_{\delta\theta=0} = J_r^{-1}(\mathrm{Log}\,R).}$$